

DG INTERNS HUB
CYBERSECURITY INTERSHIP

WEEK 3 ASSINGMENT
ADVANCED CYBERSECURITY
PRACTICAL

**Network Security Assessment, Traffic
Investigation & Hardening**

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1. Introduction

This practical demonstrates a controlled cybersecurity workflow: discover hosts, enumerate services, inspect network traffic, identify security findings, harden the environment and verify the changes.

2. Objectives

The objectives are network discovery, service enumeration, packet analysis, Nmap/Wireshark correlation, evidence-based vulnerability assessment and security hardening.

3. Lab Environment

Demo private lab: 192.168.56.0/24. Analyst VM: 192.168.56.10. Target VM: 192.168.56.20. These values are illustrative.

4. Network Diagram

The diagram shows an analyst VM connected to an isolated private lab network with two example target VMs.

5. Tools Used

Nmap for network/service discovery; Wireshark for packet analysis; VirtualBox/VMware for isolated VMs; diagrams.net for topology; GitHub for versioned submission.

6. Task 7 – Network Discovery

Commands: `nmap -sn` and `nmap` . Record IP addresses, open ports, protocols and services.

```
KAU LINUX [Running] - Oracle VirtualBox
File Machine View Input Devices Help

root@kali: ~
Session Actions Edit View Help

root@kali:~# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:0a:aa:00 brd ff:ff:ff:ff:ff:ff
    inet 192.168.29.1/24 brd 192.168.29.255 scope global dynamic noprefixroute eth0
        valid_lft 5500sec preferred_lft 5500sec
    inet6 2a05:2b1c:0000:7015:7506:93f4:dadd:45ac/64 scope global dynamic noprefixroute
        valid_lft 5592sec preferred_lft 5592sec
    inet6 fe80::b9ac:68a:d384:a2f3/64 scope link noprefixroute
        valid_lft forever preferred_lft forever

root@kali:~# ip route
default via 192.168.29.1 dev eth0 proto dhcp src 192.168.29.46 metric 100
192.168.29.0/24 dev eth0 proto kernel scope link src 192.168.29.46 metric 100

root@kali:~# nmap -sn 192.168.29.0/24
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 02:25 -0400
Nmap scan report for reliance.reliance (192.168.29.1)
Host is up (0.019s latency).
MAC Address: 8C:A3:99:70:79:98 (Servercom (India) Private Limited)
Nmap scan report for 192.168.29.53
Host is up (0.0015s latency).
MAC Address: 08:00:27:E7:F8:C6 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.29.150
Host is up (0.0011s latency).
MAC Address: 08:00:27:76:F7:09 (Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.29.151
Host is up (0.23s latency).
MAC Address: 9a:18b3:31fc:108:85 (AzureWave Technology)
Nmap scan report for 192.168.29.166
Host is up (0.15s latency).
MAC Address: 9C:16:7D:6E:9D:6A:3 (Intel Corporate)
Nmap scan report for 192.168.29.220
Host is up (0.19s latency).
MAC Address: 22:3f:95:31:5D:12 (unknown)
Nmap scan report for 192.168.29.221
Host is up (0.00039s latency).
MAC Address: 38:1031:C8:38:E3:A1 (Cloud Network Technology Singapore PTE.)
Nmap scan report for 192.168.29.46
Host is up.
Nmap done: 256 IP addresses (8 hosts up) scanned in 4.11 seconds
```

```
root@kali:~# nmap 192.168.29.53
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-02 02:25 -0400
Nmap scan report for 192.168.29.53
Host is up (0.00049s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell
1099/tcp  open  rmiregistry
1524/tcp  open  ingreslock
2049/tcp  open  nfs
2121/tcp  open  ccproxy-ftp
3306/tcp  open  mysql
5432/tcp  open  postgresql
5900/tcp  open  vnc
6000/tcp  open  X11
6667/tcp  open  irc
8009/tcp  open  ajp13
8180/tcp  open  unknown
MAC Address: 08:00:27:E7:F8:C6 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.03 seconds

root@kali:~#
```

7. Task 8 – Advanced Nmap

Use nmap -sV, nmap -O, nmap -sS and nmap -sU --top-ports 20 against authorized systems. Vulnerability scripts must only be used in the approved lab.

```
KALI LINUX (Running) - Oracle VM VirtualBox
File Machine View Input Devices Help

root@kali: ~
Session Actions Edit View Help
1900/udp closed upnp
4500/udp closed nat-t-like
49152/udp closed unknown
MAC Address: 08:00:27:E7:F8:C6 (Oracle VM VirtualBox virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 8.14 seconds

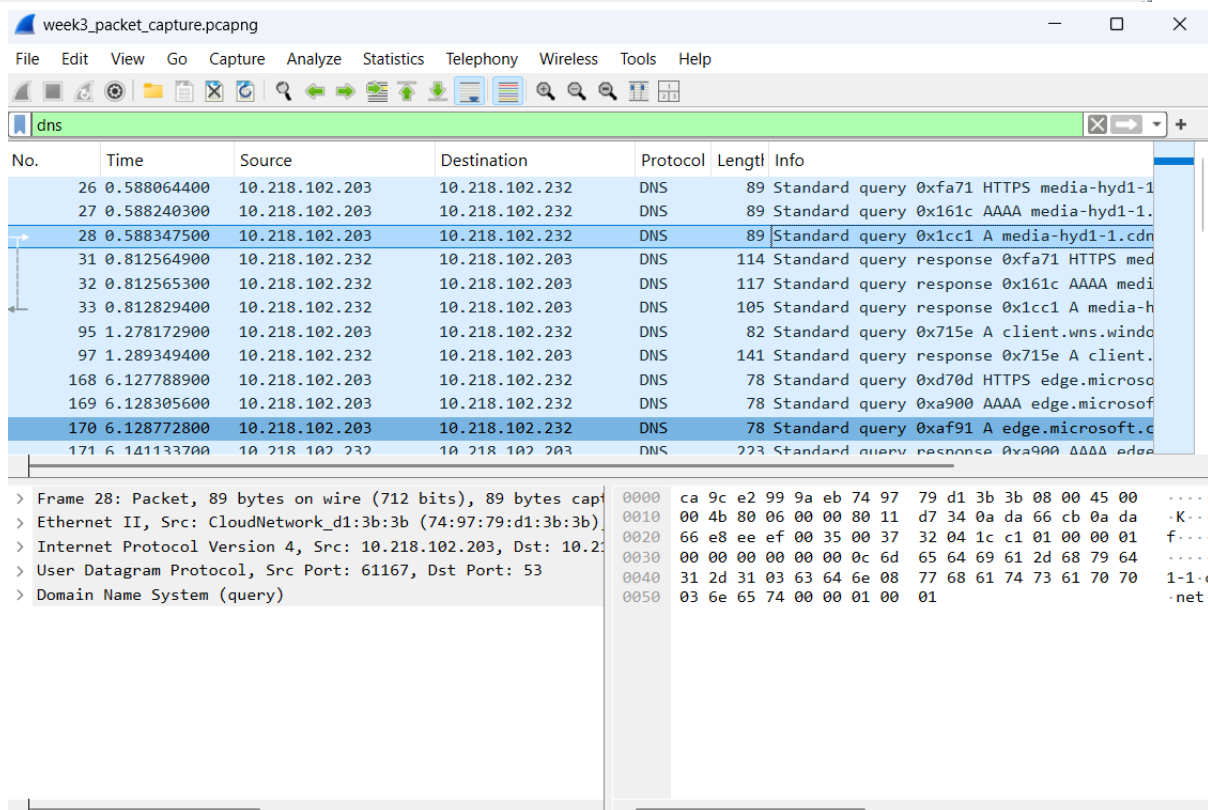
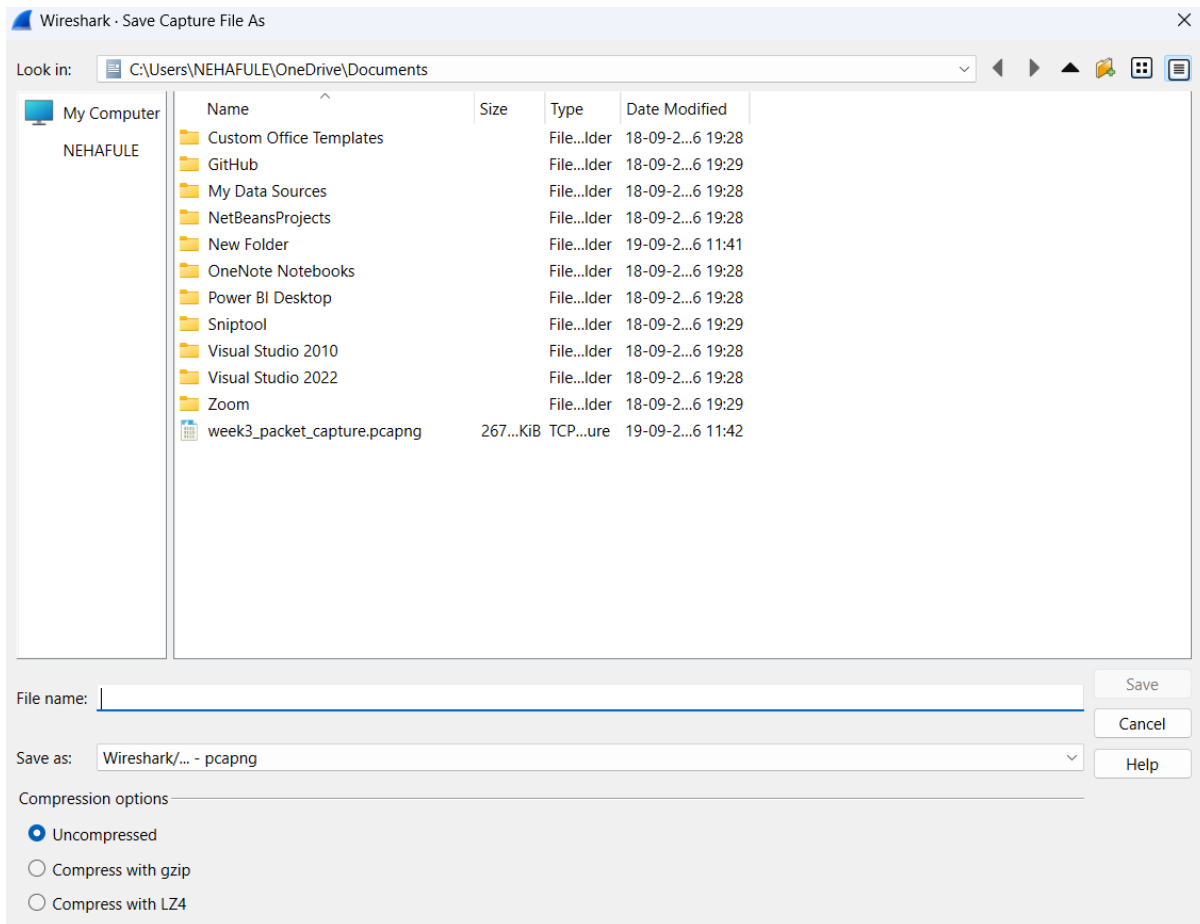
root@kali: ~# nmap --script vuln 192.168.29.53
Starting Nmap 7.91 (https://nmap.org) at 2020-09-02 02:31 -0400
Nmap scan report for 192.168.29.53
Host is up (0.00071s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
| ftp-vsftpd-backdoor:
|   VULNERABLE:
|     vsftpd version 2.3.4 backdoor
|     State: VULNERABLE (Exploitable)
|     IDS: BID:48539 CVE:CVE-2011-2523
|     vsftpd version 2.3.4 backdoor, this was reported on 2011-07-04.
|     Disclosure date: 2011-07-03
|     Exploit results:
|       Shell command: id
|       Results: uid=0(root) gid=0(root)
|     References:
|       https://github.com/rapid7/metasploit-framework/blob/master/modules/exploits/unix/ftp/vsftpd_234_backdoor.rb
|       http://scaryfastsecurity.blogspot.com/2012/07/alert-vsftpd-download-backdoored.html
|       https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2011-2523
|       https://www.securityfocus.com/bid/48539
| 22/tcp    open  ssh
| 23/tcp    open  telnet
| 25/tcp    open  smtp
| ssl-dh-params:
|   VULNERABLE:
|     Anonymous Diffie-Hellman Key Exchange MITM Vulnerability
|     State: VULNERABLE
|     Transport Layer Security (TLS) services that use anonymous
|     Diffie-Hellman key exchange only provide protection against passive
|     eavesdropping, and are vulnerable to active man-in-the-middle attacks
|     which could completely compromise the confidentiality and integrity
|     of any data exchanged over the resulting session.
|     Check results:
|       ANONYMOUS DH GROUP 1
|       Cipher Suite: TLS_DH_anon_EXPORT_WITH_DES40_CBC_SHA
|       Modulus Type: Safe prime
|       Modulus Source: Unknown/Custom-generated
|       Modulus Length: 512
|       Generator length: 8
|       Public Key Length: 512
|     References:
|       https://www.ietf.org/rfc/rfc2246.txt
| Transport Layer Security (TLS) Protocol DHE_EXPORT Ciphers Downgrade MITM (Logjam)
|   State: VULNERABLE
|   IDS: BID:74733 CVE:CVE-2015-4000
|   The Transport Layer Security (TLS) protocol contains a flaw that is
|   triggered when handling Diffie-Hellman key exchanges defined with
|   the DHE_EXPORT cipher. This may allow a man-in-the-middle attacker
|   to downgrade the security of a TLS session to 512-bit export-grade
|   cryptography, which is significantly weaker, allowing the attacker
|   to more easily break the encryption and monitor or tamper with
|   the encrypted stream.
|   Disclosure date: 2015-5-19
|   Check results:
|     EXPORT-GRADE DH GROUP 1
|     Cipher Suite: TLS_DHE_RSA_EXPORT_WITH_DES40_CBC_SHA
|     Modulus Type: Safe prime
|     Modulus Source: Unknown/Custom-generated
|     Modulus Length: 512
|     Generator length: 8
|     Public Key Length: 512
|   References:
|     https://weakdh.org
|     https://www.securityfocus.com/bid/74733
|     https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2015-4000
| Diffie-Hellman Key Exchange Insufficient Group Strength
|   State: VULNERABLE
|   Transport Layer Security (TLS) services that use Diffie-Hellman groups
|   of insufficient strength, especially those using one of a few commonly
|   shared groups, may be susceptible to passive eavesdropping attacks.
|   Check results:
|     WEAK DH GROUP 1
|     Cipher Suite: TLS_DHE_RSA_WITH_3DES_EDE_CBC_SHA
|     Modulus Type: Safe prime
|     Modulus Source: postfix builtin
|     Modulus Length: 1024
|     Generator length: 8
|     Public Key Length: 1024
|   References:
|     https://weakdh.org
| sslv2-drown: ERROR: Script execution failed (use -d to debug)
| ssl-poodle:
|   VULNERABLE:
|     SSL POODLE Information Leak
|     State: VULNERABLE
|     IDS: BID:76576 CVE:CVE-2014-3566
|     The SSL protocol 3.0, as used in OpenSSL through 1.0.1i and other
|     products, uses nondeterministic CBC padding, which makes it easier
|     for man-in-the-middle attackers to obtain cleartext data via a
```

```
KALI LINUX (Running) - Oracle VM VirtualBox
File Machine View Input Devices Help

root@kali: ~
Session Actions Edit View Help
References:
| https://www.ietf.org/rfc/rfc2246.txt
| Transport Layer Security (TLS) Protocol DHE_EXPORT Ciphers Downgrade MITM (Logjam)
|   State: VULNERABLE
|   IDS: BID:74733 CVE:CVE-2015-4000
|   The Transport Layer Security (TLS) protocol contains a flaw that is
|   triggered when handling Diffie-Hellman key exchanges defined with
|   the DHE_EXPORT cipher. This may allow a man-in-the-middle attacker
|   to downgrade the security of a TLS session to 512-bit export-grade
|   cryptography, which is significantly weaker, allowing the attacker
|   to more easily break the encryption and monitor or tamper with
|   the encrypted stream.
|   Disclosure date: 2015-5-19
|   Check results:
|     EXPORT-GRADE DH GROUP 1
|     Cipher Suite: TLS_DHE_RSA_EXPORT_WITH_DES40_CBC_SHA
|     Modulus Type: Safe prime
|     Modulus Source: Unknown/Custom-generated
|     Modulus Length: 512
|     Generator length: 8
|     Public Key Length: 512
|   References:
|     https://weakdh.org
|     https://www.securityfocus.com/bid/74733
|     https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2015-4000
| Diffie-Hellman Key Exchange Insufficient Group Strength
|   State: VULNERABLE
|   Transport Layer Security (TLS) services that use Diffie-Hellman groups
|   of insufficient strength, especially those using one of a few commonly
|   shared groups, may be susceptible to passive eavesdropping attacks.
|   Check results:
|     WEAK DH GROUP 1
|     Cipher Suite: TLS_DHE_RSA_WITH_3DES_EDE_CBC_SHA
|     Modulus Type: Safe prime
|     Modulus Source: postfix builtin
|     Modulus Length: 1024
|     Generator length: 8
|     Public Key Length: 1024
|   References:
|     https://weakdh.org
| sslv2-drown: ERROR: Script execution failed (use -d to debug)
| ssl-poodle:
|   VULNERABLE:
|     SSL POODLE Information Leak
|     State: VULNERABLE
|     IDS: BID:76576 CVE:CVE-2014-3566
|     The SSL protocol 3.0, as used in OpenSSL through 1.0.1i and other
|     products, uses nondeterministic CBC padding, which makes it easier
|     for man-in-the-middle attackers to obtain cleartext data via a
```

8. Task 9 – Wireshark

Capture authorized lab traffic and investigate DNS, TCP, UDP, ICMP, ARP and HTTP where present. Useful filters include dns, tcp, udp, icmp, arp and tcp.flags.syn == 1.



week3_packet_capture.pcapng

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

dns

No.	Time	Source	Destination	Protocol	Length	Info
26	0.588064400	10.218.102.203	10.218.102.232	DNS	89	Standard query 0xfa71 HTTPS media-hyd1-1
27	0.588240300	10.218.102.203	10.218.102.232	DNS	89	Standard query 0x161c AAAA media-hyd1-1.
28	0.588347500	10.218.102.203	10.218.102.232	DNS	89	Standard query 0x1cc1 A media-hyd1-1.cdn
31	0.812564900	10.218.102.232	10.218.102.203	DNS	114	Standard query response 0xfa71 HTTPS med
32	0.812565300	10.218.102.232	10.218.102.203	DNS	117	Standard query response 0x161c AAAA medi
33	0.812829400	10.218.102.232	10.218.102.203	DNS	105	Standard query response 0x1cc1 A media-h
95	1.278172900	10.218.102.203	10.218.102.232	DNS	82	Standard query 0x715e A client.wns.windo
97	1.289349400	10.218.102.232	10.218.102.203	DNS	141	Standard query response 0x715e A client.
168	6.127788900	10.218.102.203	10.218.102.232	DNS	78	Standard query 0xd70d HTTPS edge.microsof
169	6.128305600	10.218.102.203	10.218.102.232	DNS	78	Standard query 0xa900 AAAA edge.microsof
170	6.128772800	10.218.102.203	10.218.102.232	DNS	78	Standard query 0xaf91 A edge.microsoft.c
171	6.141133700	10.218.102.232	10.218.102.203	DNS	223	Standard query response 0xa900 AAAA edge

> Frame 32: Packet, 117 bytes on wire (936 bits), 117 bytes captured (936 bits) on interface 0

> Ethernet II, Src: ca:9c:e2:99:9a:eb (ca:9c:e2:99:9a:eb), Dst: 10.218.102.203

> Internet Protocol Version 4, Src: 10.218.102.232, Dst: 10.218.102.203

> User Datagram Protocol, Src Port: 53, Dst Port: 62178

> Domain Name System (response)

0000 74 97 79 d1 3b 3b ca 9c e2 99 9a eb 08 00 45 00 t.y.

0010 00 67 39 4a 40 00 00 11 1d d5 0a da 66 e8 0a da g9J0

0020 66 cb 00 35 f2 e2 00 53 ed 42 16 1c 81 80 00 01 f..5

0030 00 01 00 00 00 00 0c 6d 65 64 69 61 2d 68 79 64

0040 31 2d 31 03 63 64 6e 08 77 68 61 74 73 61 70 70 1-1.C

0050 03 6e 65 74 00 00 1c 00 01 c0 0c 00 1c 00 01 00 -net

0060 00 07 56 00 10 2a 03 28 80 f2 85 00 d0 fa ce b0 ..V-

0070 0c 00 00 01 67E

Bytes 87-88: Class (dns.qry.class) | Packets: 579 · Displayed: 38 (6.6%) | Profile: Default

9. Task 10 – Nmap + Wireshark

Run an authorized SYN scan while capturing traffic. Compare SYN/SYN-ACK for open ports, SYN/RST for closed ports and non-response where filtering may exist.

KALI LINUX [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

1 2 3 4

*eth0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.flags.syn

No.	Time	Source	Destination	Protocol	Length	Info
3393	295.416142679	10.180.135.195	10.180.135.212	TCP	66	59986 → 1524 [ACK] Seq=1 Ack=24 Win=64256 Len=0 TSval=2579565994 TSecr=51365
3394	295.418923641	10.180.135.195	10.180.135.212	TCP	66	59986 → 1524 [FIN, ACK] Seq=1 Ack=24 Win=64256 Len=0 TSval=2579565995 TSecr=51365
3395	295.412662460	10.180.135.212	10.180.135.195	TCP	71	1524 → 59986 [PSH, ACK] Seq=24 Ack=2 Win=5792 Len=5 TSval=51365 TSecr=2579565995
3396	295.412685923	10.180.135.195	10.180.135.212	TCP	54	59986 → 1524 [RST] Seq=2 Win=0 Len=0
3397	296.359785627	10.180.135.212	10.180.135.195	IRC	121	Response (ERROR)
3398	296.359817783	10.180.135.195	10.180.135.212	TCP	54	54082 → 6667 [RST] Seq=2 Win=0 Len=0
3399	298.229851874	10.180.135.212	10.180.135.195	TCP	74	[TCP Retransmission] 80 → 52942 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=51645 TSecr=2579565788 WS=32
3400	298.229897857	10.180.135.195	10.180.135.212	TCP	66	[TCP Dup ACK 3286#1] 52942 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2579568814 TSecr=51345
3406	301.223102889	10.180.135.195	10.180.135.212	DNS	98	Standard query 0x0006 TXT version.bind
3407	301.224277365	10.180.135.212	10.180.135.195	TCP	66	53 → 39666 [ACK] Seq=1 Ack=33 Win=5792 Len=0 TSval=51938 TSecr=2579571807
3408	301.224628734	10.180.135.195	10.180.135.212	HTTP	84	GET / HTTP/1.0
3409	301.225133793	10.180.135.195	10.180.135.212	Portmap	110	V104316 proc=0 Call (Reply In 3426)
3410	301.225827318	10.180.135.212	10.180.135.195	TCP	66	80 → 52942 [ACK] Seq=1 Ack=19 Win=5792 Len=0 TSval=51938 TSecr=2579571808
3411	301.225827622	10.180.135.212	10.180.135.195	TCP	66	111 → 35840 [ACK] Seq=1 Ack=45 Win=5792 Len=0 TSval=51938 TSecr=2579571809
3412	301.226325438	10.180.135.195	10.180.135.212	NBSS	84	NBSS Continuation Message
3413	301.226684733	10.180.135.195	10.180.135.212	SMB	234	Negotiate Protocol Request
3414	301.227238793	10.180.135.212	10.180.135.195	TCP	66	139 → 53854 [ACK] Seq=1 Ack=19 Win=5792 Len=0 TSval=51938 TSecr=2579571810
3415	301.227658317	10.180.135.212	10.180.135.195	TCP	66	445 → 60402 [ACK] Seq=1 Ack=169 Win=6880 Len=0 TSval=51938 TSecr=2579571811
3416	301.227766573	10.180.135.195	10.180.135.212	Rlogin	98	Start Handshake
3417	301.228132936	10.180.135.195	10.180.135.212	RMI	73	JRMI, Version: 2, StreamProtocol

KALI LINUX [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

1 2 3 4

*eth0

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tcp.flags.reset

No.	Time	Source	Destination	Protocol	Length	Info
3393	295.416142679	10.180.135.195	10.180.135.212	TCP	66	59986 → 1524 [ACK] Seq=1 Ack=24 Win=64256 Len=0 TSval=2579565994 TSecr=51365
3394	295.418923641	10.180.135.195	10.180.135.212	TCP	66	59986 → 1524 [FIN, ACK] Seq=1 Ack=24 Win=64256 Len=0 TSval=2579565995 TSecr=51365
3395	295.412662460	10.180.135.212	10.180.135.195	TCP	71	1524 → 59986 [PSH, ACK] Seq=24 Ack=2 Win=5792 Len=5 TSval=51365 TSecr=2579565995
3396	295.412685923	10.180.135.195	10.180.135.212	TCP	54	59986 → 1524 [RST] Seq=2 Win=0 Len=0
3397	296.359785627	10.180.135.212	10.180.135.195	IRC	121	Response (ERROR)
3398	296.359817783	10.180.135.195	10.180.135.212	TCP	54	54082 → 6667 [RST] Seq=2 Win=0 Len=0
3399	298.229851874	10.180.135.212	10.180.135.195	TCP	74	[TCP Retransmission] 80 → 52942 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM TSval=51645 TSecr=2579565788 WS=32
3400	298.229897857	10.180.135.195	10.180.135.212	TCP	66	[TCP Dup ACK 3286#1] 52942 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2579568814 TSecr=51345
3406	301.223102889	10.180.135.195	10.180.135.212	DNS	98	Standard query 0x0006 TXT version.bind
3407	301.224277365	10.180.135.212	10.180.135.195	TCP	66	53 → 39666 [ACK] Seq=1 Ack=33 Win=5792 Len=0 TSval=51938 TSecr=2579571807
3408	301.224628734	10.180.135.195	10.180.135.212	HTTP	84	GET / HTTP/1.0
3409	301.225133793	10.180.135.195	10.180.135.212	Portmap	110	V104316 proc=0 Call (Reply In 3426)
3410	301.225827318	10.180.135.212	10.180.135.195	TCP	66	80 → 52942 [ACK] Seq=1 Ack=19 Win=5792 Len=0 TSval=51938 TSecr=2579571808
3411	301.225827622	10.180.135.212	10.180.135.195	TCP	66	111 → 35840 [ACK] Seq=1 Ack=45 Win=5792 Len=0 TSval=51938 TSecr=2579571809
3412	301.226325438	10.180.135.195	10.180.135.212	NBSS	84	NBSS Continuation Message
3413	301.226684733	10.180.135.195	10.180.135.212	SMB	234	Negotiate Protocol Request
3414	301.227238793	10.180.135.212	10.180.135.195	TCP	66	139 → 53854 [ACK] Seq=1 Ack=19 Win=5792 Len=0 TSval=51938 TSecr=2579571810
3415	301.227658317	10.180.135.212	10.180.135.195	TCP	66	445 → 60402 [ACK] Seq=1 Ack=169 Win=6880 Len=0 TSval=51938 TSecr=2579571811
3416	301.227766573	10.180.135.195	10.180.135.212	Rlogin	98	Start Handshake
3417	301.228132936	10.180.135.195	10.180.135.212	RMI	73	JRMI, Version: 2, StreamProtocol

DEMO ONLY — replace with actual authorized-lab evidence.

10. Task 11 – Advanced Traffic Investigation

Review top hosts, DNS activity, TCP/UDP behavior and repeated connection attempts. Suspicious-looking activity should be validated with additional evidence rather than automatically labeled malicious.

DEMO ONLY — replace with actual authorized-lab evidence.

11. Task 12 – Vulnerability Assessment

Demo findings: unnecessary exposed service (Medium), broad service exposure (Medium), outdated software requiring patch review (High if confirmed), weak service configuration (Medium), and insufficient network restriction (Medium). Actual severity must be based on real evidence.

12. Task 13 – Security Hardening

Example improvements: disable unnecessary service, restrict firewall exposure, update software, limit administrative access and verify with a second scan.

Selected Ports — Before / After Verification

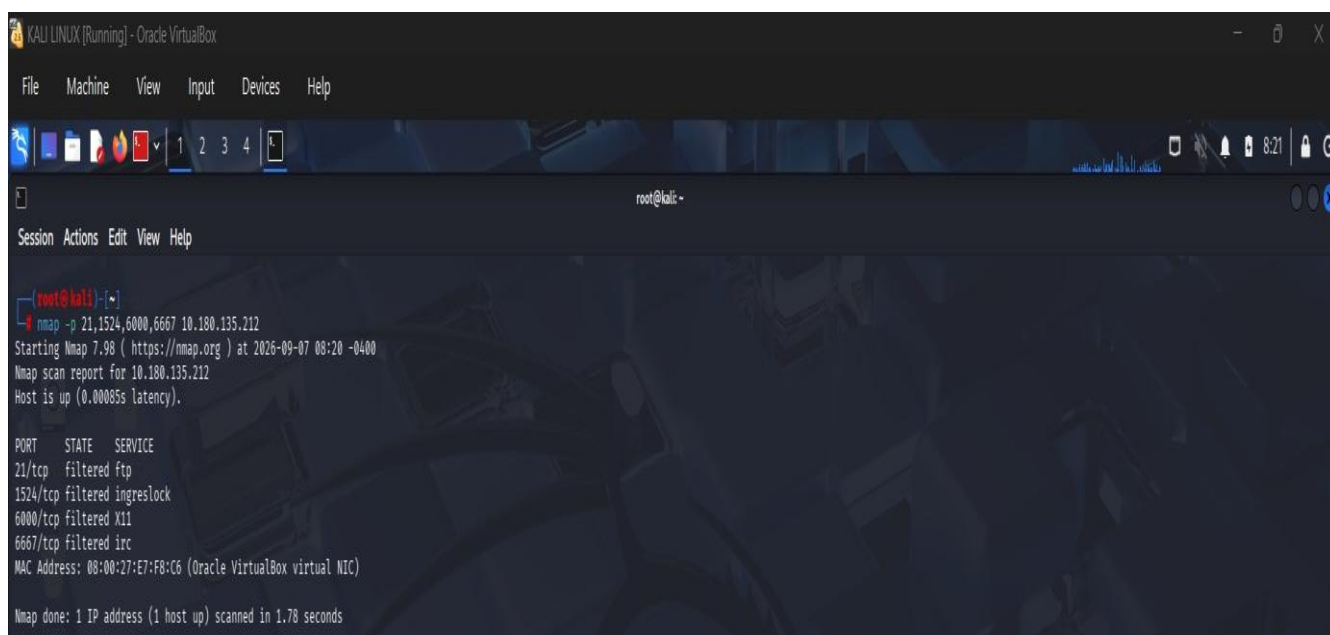
Before: 192.168.29.53 | After: 10.180.135.212

Port	Service	Before	After	Risk	Interpretation
21	FTP	Open	Filtered	Critical	Exposure reduced; fil
1524	Ingreslock / bindshell	Open	Filtered	Critical	Exposure reduced; fil
6000	X11	Open	Filtered	Medium	Exposure reduced; fil
6667	IRC	Open	Filtered	Critical	Exposure reduced; fil

Note: IP address changed between captures; treat this as a lab-state comparison, not exact same-IP proof.

13. Task 14 – Final Mini Assessment

Summarize discovery, traffic analysis, security findings, hardening actions and verification. Include five recommendations and attach real evidence in the final practical.



```
KALI LINUX [Running] - Oracle VirtualBox
File Machine View Input Devices Help

root@kali: ~
Session Actions Edit View Help

root@kali:~# nmap -p 21,1524,6000,6667 10.180.135.212
Starting Nmap 7.98 ( https://nmap.org ) at 2026-09-07 08:20 -0400
Nmap scan report for 10.180.135.212
Host is up (0.00085s latency).

PORT      STATE SERVICE
21/tcp    filtered ftp
1524/tcp   filtered ingreslock
6000/tcp   filtered X11
6667/tcp   filtered irc
MAC Address: 08:00:27:E7:F8:C6 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.78 seconds
```